Data

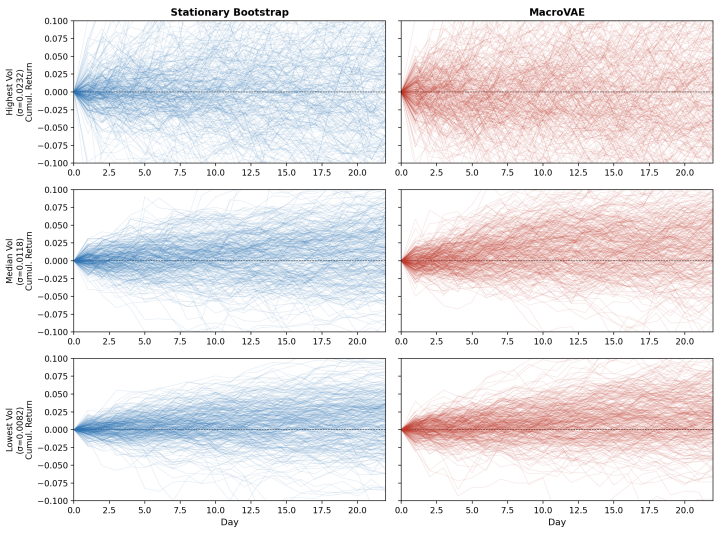
[M, N, D] for m samples and n stocks and d days.
for stocknet M = 2048, N = 81, D = 22

Note:

1. W1(Wasserstein-1 Distance): computed on cross-sectional.
2. Mean/Std/Skew/Kurt: computed per asset and aggregated as mean(std).
3. Cov: Pull as [M*D, N] and Frobenius norm of correlation matrix on each sample
4. ES: Cumulative return on temporal, get ES per asset, aggregated as mean(std).

Results

Method	W1↓	Cov↓	ES↓	Mean↓	Std↓	Skew↓	Kurt↓	LinUnpred↓	VolClust↓	CoarseFine↓
Stationary Bootstrap	0.0000	0.0160	-0.063±0.10	+0.254±0.84	+0.001±0.01	+0.034±0.55	-0.019±0.05	-0.007	+0.001	-0.212
MacroVAE	0.0021	0.6190	-0.300±0.21	-0.231±2.42	+0.150±0.07	-0.743±4.44	-0.483±0.34	-0.067	-0.072	-0.719



Simulated cumulative return paths (300 of 2048 scenarios) for the highest-, median-, and lowest-volatility stocks.

Appendix: Metrics

Let $r_{m,n,t}$ denote the return of asset n at day t in scenario m. All metrics are computed per scenario per asset on the D-day series, then averaged over M scenarios and N assets. Reported as relative error $(generated - reference)/|reference|$.

- **Lin. Unpred. (Linear Unpredictability)**: serial autocorrelation of the return series $\{r_{m,n,t}\}_{t=1}^{D}$ at lags $k = 1, \ldots, 10$, averaged over lags, assets, and scenarios.
- **Vol Clust. (Volatility Clustering)**: serial autocorrelation of the absolute return series $\{|r_{m,n,t}|\}_{t=1}^{D}$ at lags $k = 1, \ldots, 10$, averaged over lags, assets, and scenarios.
- **Coarse-Fine**: cross-correlation between coarse and fine volatility measures at lag k, with rolling window $\tau = 5$ days.

$$\text{coarse}_t = \left|\sum_{s=t-\tau+1}^{t} r_s\right|, \quad \text{fine}_t = \sum_{s=t-\tau+1}^{t} |r_s|$$

$$\overline{\mathrm{CF}} = \frac{1}{KN} \sum_{k=1}^{K} \sum_{n=1}^{N} \mathrm{Corr}_m(\mathrm{coarse}_{t+k},\, \mathrm{fine}_t \mid n)$$

where Corr_m averages the per-scenario correlation over M scenarios.